

House Price Prediction using Linear Regression

**Main Crafts AI & Machine Learning
Internship – Task 1**

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1. Introduction

The objective of this project is to build and evaluate a Machine Learning model to predict house prices using the California Housing dataset. This project demonstrates the complete Machine Learning workflow, including data loading, exploratory data analysis, preprocessing, model training, evaluation, and visualization.

House price prediction is an important real-world problem that helps in understanding market trends and supports decision-making in real estate and finance sectors. In this project, Linear Regression is used as a baseline regression model to predict the median house value.

2. Dataset Description

The California Housing dataset is a real-world dataset provided by the Scikit-learn library. It contains housing information collected from different regions of California.

Dataset Features:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

Target Variable:

- **Median House Value (MedHouseVal)**

The dataset contains only numerical values and does not include missing values, making it suitable for regression analysis.

3. Exploratory Data Analysis (EDA)

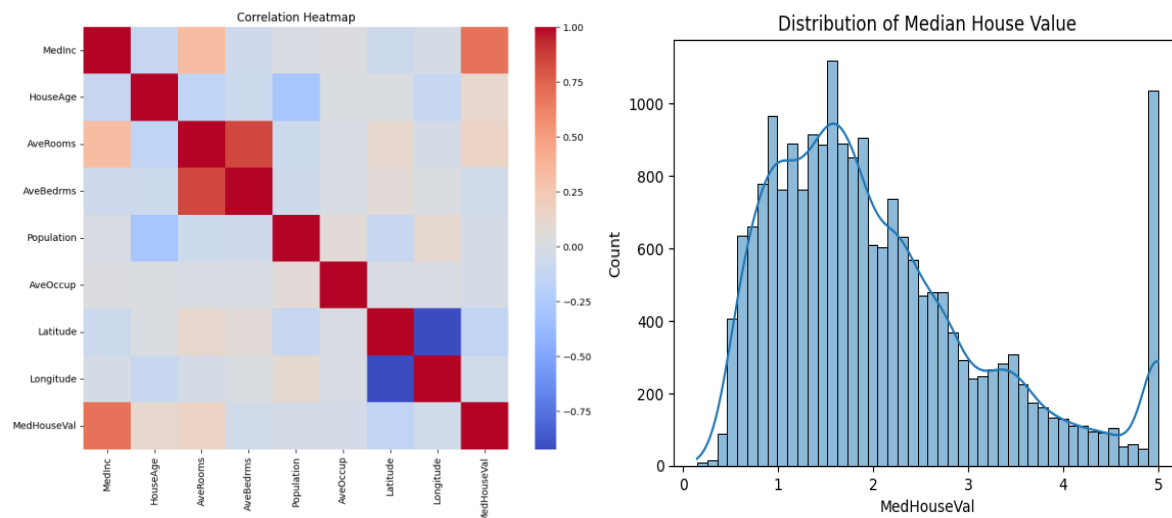
Exploratory Data Analysis was performed to understand the dataset structure, feature distributions, and relationships between variables.

EDA Steps Performed:

- Checked dataset information and statistical summary
- Analyzed feature correlations using a heatmap
- Visualized the distribution of the target variable

Observations:

- Median Income shows a strong positive correlation with house prices
- Some features have weak correlations, indicating limited impact
- Target variable follows a slightly right-skewed distribution



4. Data Preprocessing

The dataset was prepared for model training by separating the independent variables (features) and the dependent variable (target).

The data was split into:

- **Training Set:** 80%
- **Testing Set:** 20%

This ensures that the model is evaluated on unseen data to check its generalization capability.

5. Model Training

A **Linear Regression** model was trained using the training dataset. Linear Regression is a supervised learning algorithm used to predict continuous values by finding the best-fit linear relationship between input features and the target variable.

The model was trained using Scikit-learn's Linear Regression class.

6. Model Evaluation

The trained model was evaluated using the following performance metrics:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between actual and predicted values
- **Root Mean Squared Error (RMSE):** Penalizes larger errors more heavily
- **R² Score:** Indicates how well the model explains the variance in the data

These metrics help assess the accuracy and reliability of the model.

Mean Absolute Error (MAE): 0.5332001304956565

Root Mean Squared Error (RMSE): 0.745581383012776

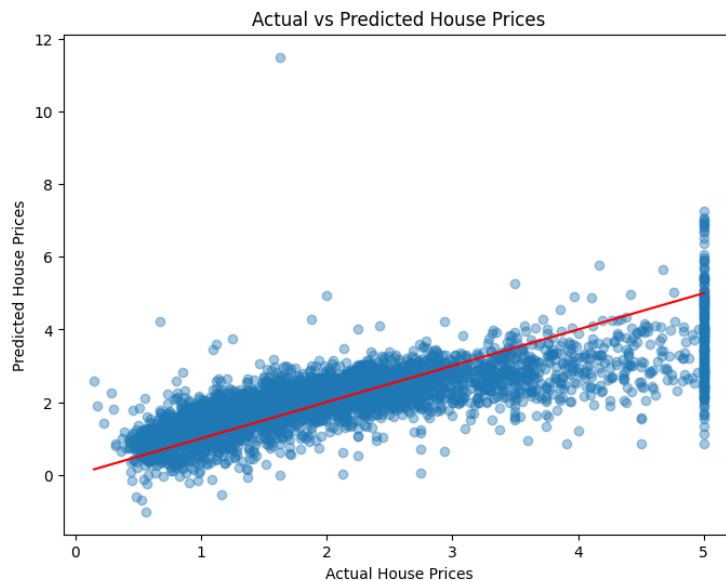
R² Score: 0.5757877060324512

7. Results & Visualization

The model's predictions were compared with actual house prices using a scatter plot.

Observation:

- Most points are aligned near the diagonal line
- Indicates reasonable prediction performance
- Some deviation exists due to the simplicity of Linear Regression



8. Conclusion & Future Scope

In this project, a Linear Regression model was successfully implemented to predict house prices using the California Housing dataset. The model demonstrated reasonable performance and helped understand the end-to-end Machine Learning pipeline.

Future Improvements:

- Apply feature scaling
- Use advanced models like Random Forest or XGBoost
- Perform hyperparameter tuning
- Use additional datasets for better generalization